A New Family of Operators on Cellular Automata

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Abstract

We describe a cellular automaton whose cells hold not states but *rules*, and an operator P_σ that rewrites each rule by acting on its own truth table, $\mathrm{bit}[\sigma(k)] := \neg\,\mathrm{bit}[k]$ for σ a permutation of the eight neighbourhoods (Def. 1). This gives a finite, exhaustively classifiable family of $8! = 40{,}320$ operators—a vanishing 0.24% of the general *writings* (Obs. 1).

The classification is exact and machine-checked: P_σ has a fixed rule if and only if every cycle of σ is even (Thm. 1), and every fixed rule is *balanced*, at Langton's $\lambda = 1/2$ (Thm. 2). A census of all 40,320 makes this a measured law: any odd cycle forbids collapse, the all-even class splits, and every genotype/phenotype regime is reachable by a bijection (Table 1).

Balance is exact, but edge-of-chaos *behaviour* is a separate question: only the coprime 8-cycles sustain a high-diversity phase (Emp. find. 1, Fig. 6). That phase has no critical point but a finite-width *littoral edge* (Emp. find. 2, 3; Fig. 12), whose ordered and chaotic domains meet at fractal walls where the genotype is rewritten fastest (Emp. find. 4, Fig. 14). The family is flat under its own composition (Obs. 4); yet pairing a propagator with a construction step—a *river*—or braiding complementary operators nonetheless manufactures sustained structure (Figs. 9, 11).

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The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Invert that arrangement: let each cell hold a *rule*, and let an operator evolve the field of rules. An elementary cellular automaton (ECA) rule is an eight-bit truth table $\mathrm{bit}[k]$, with k ranging over the eight three-cell neighbourhoods. We study the operator P_σ, parameterised by a permutation $\sigma \in S_8$ that acts on the rule's own table. Writing a rule as a map $g\colon \{0,\ldots,7\} \to \{0,1\}$, the operator sends g to $P_\sigma g$ defined by

$$(P_\sigma\, g)(j) \;=\; \neg\, g\Big(\sigma^{-1}(j)\Big), \qquad \text{equivalently} \qquad \mathrm{bit}[\sigma(k)] := \neg\,\mathrm{bit}[k]. \tag{1}$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of $8! = 40{,}320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified.

Definition 1 (Writings, propagators, and permutations)**.** *A* writing *is any function $w\colon \{0,\ldots,7\} \to \{0,\ldots,7\}$ on the truth-table positions, and its* propagator P_w *complements each source bit and writes it to the target $w(k)$:* $\mathrm{bit}[w(k)] := \neg\,\mathrm{bit}[k]$*. When w is a bijection—a permutation $\sigma \in S_8$—each target is written exactly once, σ^{-1} exists, and P_σ takes the closed form* (1)*; these are the* permutation-propagators*. A non-injective writing has* collisions *(two sources sharing a target, so the simultaneous rewrite is order-dependent) and* gaps *(a position written by none).*

Observation 1: The permutation core is a vanishing fraction

Permutations are only the bijective core of a far larger class: there are $8! = 40{,}320$ permutations but $8^8 = 16{,}777{,}216$ writings, so the permutations are 0.24% of the whole.

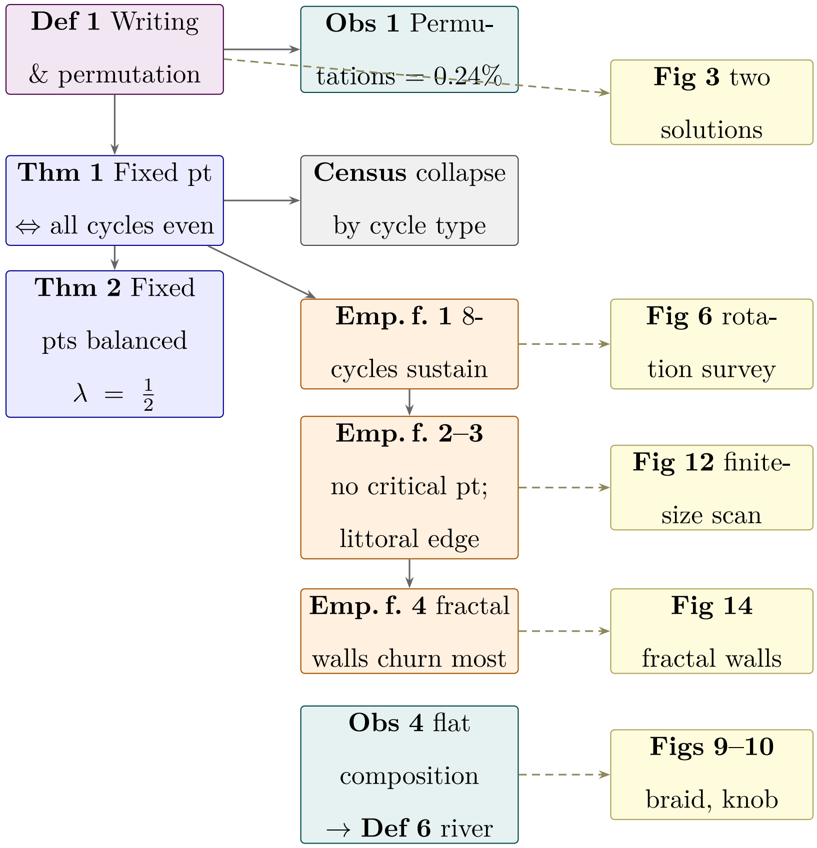

Figure 1

Map of the paper by box. *Solid arrows carry logical support;* dashed *arrows link each claim to the* ***figure*** *that visualises it (yellow)—the argument rests on what can be seen. Colour encodes type: definition (violet), theorem (blue), observation (teal), empirical finding (amber), census (grey), figure (yellow). Only the principal boxes are drawn; the paper carries some two dozen typed boxes in all, plus exposition boxes for the connecting prose. Anything not in a box is droppable scaffolding.*

The cycle-structure algebra of *The Classification* is available only on this core. The historical operator [0 0 1 2 3 4 5 6] (Figure 3) is itself a non-bijective writing—bits 0 and 1 both write to position 0, and position 7 is never written—and its behaviour ranges widely between its two standard-order forms.

Box 1 works through one member of the family bit by bit.

Worked example 1: A propagator by hand

Write an ECA rule in Wolfram's standard neighbourhood order:

neighbourhood	111	110	101	100	011	010	001	000
Rule 110	0	1	1	0	1	1	1	0

Thus Rule 110 is the byte 01101110. Let $\sigma(k) = k+2 \pmod 8$: each source bit is complemented and written two positions further along. Carrying out the eight writes gives

$$P_{+2}(01101110) = 01100100,$$

which is Rule 100. This changes what the cell does to its black-and-white phenotype. For example, neighbourhood 011 maps to 1 under Rule 110 but to 0 under Rule 100. The propagator has not directly flipped a phenotype cell; it has changed the truth table that the cell will use.

The bare operator acts on one eight-bit rule; in the spatial experiments it is placed inside a two-layer automaton.

Definition 2 (Genotype and phenotype)**.** *Each cell carries an eight-bit rule. The field of rules—the* genotype*—governs a field of black-and-white states—the* phenotype*—and neighbouring rules also interact. One complete local update (Box 2) evolves both layers; this is the computation behind the space–time diagrams in Figure 2.*

Worked example 2: One site, one complete MetaCA step

At time t, site x holds an eight-bit rule $g_x(t)$ and a phenotype bit $X_x(t)$. One ordinary propagator step has three parts.

1. **Express the current rule.** The phenotype neighbourhood is evaluated by the rule at its centre:

$$X_x(t+1) = g_x(t)\big(X_{x-1}(t), X_x(t), X_{x+1}(t)\big).$$

2. **Blend neighbouring rules.** For each truth-table position j, inspect the three genotype bits $\big(g_{x-1}(j), g_x(j), g_{x+1}(j)\big)$. If the two outer bits agree, copy their value. Thus $(0,1,0)$ blends to 0. If they disagree, the centre rule decides by treating the triple as one of its ECA neighbourhoods. The eight coordinate-wise decisions form an intermediate rule

$$h_x = B\big(g_{x-1}(t), g_x(t), g_{x+1}(t)\big).$$

3. **Propagate the blended rule.** Apply the selected wiring to that intermediate byte:

$$g_x(t+1) = P_\sigma(h_x).$$

The ordinary runs apply P_σ once; the historical Figure 8 accident applied its off-by-one propagator twice.

The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. That 2015 paper found a single member of this family, encountered through a programming error and not recognised as one of many; we now give the family an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection × complement, collapsing 256 rules to 88 classes) (Schaller & Svozil, 2025); it is not S_8, and the negation-carrying, σ-parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν-CA (Dennunzio et al., 2011)) and rules that change during evolution (rule-changing or programmable CA (Mori et al., 1998)). A referee should note the trap in the first: a ν-CA drawing rules from a finite set is equivalent to a uniform CA on the product alphabet (Dennunzio et al., 2011), so the substrate is not where novelty lives. It lives in the fixed-point structure of the operator, which is what the rest of this paper is about.

The Classification

Fixed points exist iff every cycle is even

Theorem 1. *P_σ has a fixed point (a fixed rule) if and only if every cycle of σ has even length. Since a one-cycle has odd length, this already requires σ to be fixed-point-free.*

A fixed rule is a g invariant under (1), i.e. a σ-equivariant Boolean colouring with $g(\sigma k) = \neg\, g(k)$. Follow a cycle of length L: equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no such colouring, which is why even cycles are required throughout.

Box 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.

Worked example 3: Even and odd cycles

For offset +2 on eight truth-table positions,

$$\sigma = (0\ 2\ 4\ 6)(1\ 3\ 5\ 7).$$

A fixed rule must change colour at every arrow because $g(\sigma k) = \neg g(k)$. Around either four-cycle it can therefore read 0101 or 1010: after four negations it returns consistently to its starting bit. The two four-cycles are coloured independently, so two binary choices fix all eight truth-table entries $\mathrm{bit}[0], \ldots, \mathrm{bit}[7]$. Read back as an ECA rule number—those eight bits in

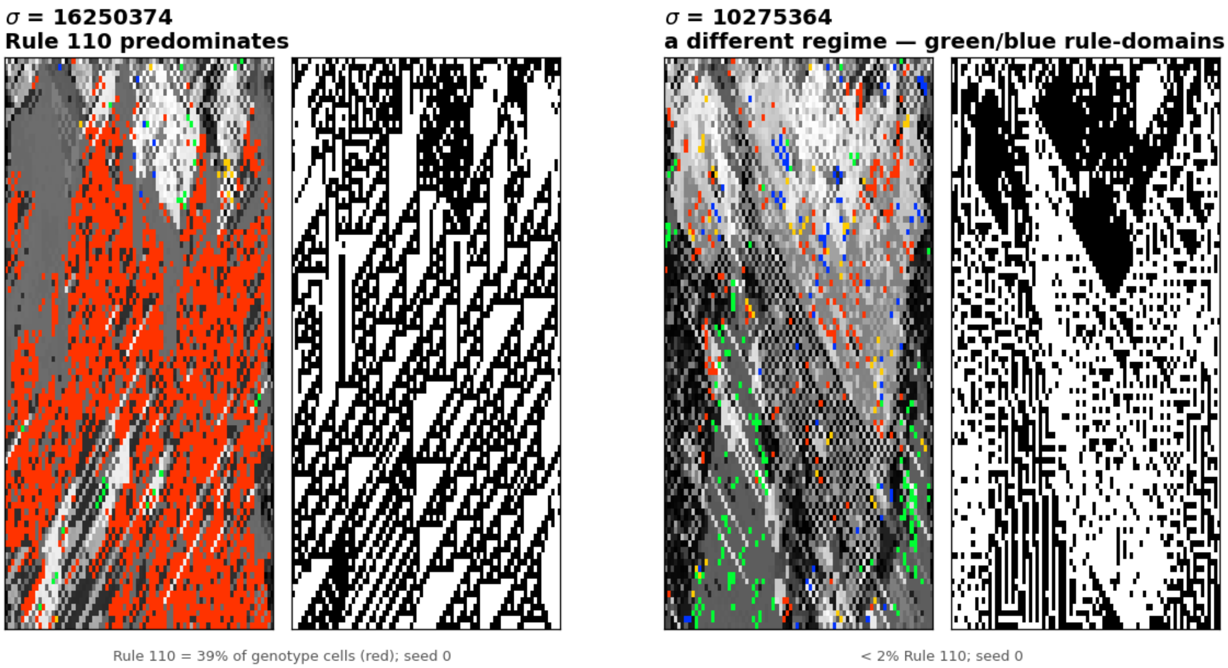

Figure 2

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ *the genotype field comes to be dominated by ECA Rule 110—the red cells, 39% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110's characteristic gliders; under* $\sigma = 10275364$ *a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine's own.*

Wolfram neighbourhood order (the table of *The Object*) taken as an integer in 0–255—the four colourings are the fixed rules

$$2^2 = 4: \quad 51,\ 102,\ 153,\ 204,$$

each with four zero-bits and four one-bits. An odd cycle admits no such colouring: after an odd number of negations the bit returns to its starting position demanding its own opposite, so no fixed rule exists. Admitting a fixed rule and settling onto one are distinct. An operator carrying an odd cycle has no fixed rule for its field to collapse onto, and its field correspondingly stays diverse (the any-odd reduced rotate+2, Figure 5, right); whether a fixed-point-bearing operator collapses onto one of its fixed rules is a separate, dynamical question (*Fixed Points Are Not Attractors*).

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena, 2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two. We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.[1]

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n-1)!!)^2$, with exponential generating function $\sum_n ((2n-1)!!)^2 x^{2n}/(2n)! = 1/\sqrt{1-x^2}$ (the "set of even cycles" construction). For S_8 this is $105^2 = 11{,}025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

[1] `DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`; `lake build` clean, no `sorry` or `axiom`.

Fixed bytes number 2^k

Corollary 1. *An all-even σ with k cycles has exactly 2^k fixed bytes; any σ with an odd cycle has none.*

Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k; the zero case is Theorem 1. This too is machine-checked.[2] Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40{,}320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Definition 3 (Langton's activity λ)**.** *λ measures a rule's* activity*: the fraction of its neighbourhood entries that map to the non-quiescent state (Langton, 1990)—for an eight-bit ECA table, the Hamming weight over eight, $\lambda = \|r\|_1/8$, from 0 (every neighbourhood dies) through $1/2$ to 1. A rule with $\lambda = 1/2$ is* balanced.

Observation 2: Balance is a truth-table property

Langton's thesis was that complex behaviour concentrates near a critical λ, shown most clearly for larger tables ($K = 4$, $N = 5$); for the two-symbol, three-neighbour ECAs Langton is explicit that λ only roughly correlates with dynamics, and later work is cautious (Mitchell et al., 1993). Theorem 2 below concerns the *value* $\lambda = 1/2$ itself—balance—which we keep separate from any empirical claim about what it predicts dynamically.

Every fixed point has $\lambda = 1/2$

Theorem 2. *Every fixed point of every operator in the family has exactly four one-bits; equivalently, Langton's activity parameter $\lambda = 4/8 = 1/2$.*

By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle

[2] `alternatingColouring_card_eq_two_pow_cycleType_card`.

exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$. Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions. The statement is machine-checked for Fin 8.[3]

We are deliberate about the standing of this result. Langton's critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)'s specific genetic-algorithm clustering claim, not the definition of λ, which survived as one axis among others such as Wuensche's Z (Wuensche, 1999). There is a folklore intuition for why 1/2 is special — binomial sampling variance $\lambda(1-\lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule's dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value 1/2 is *forced* as an algebraic matter, because any alternation-under-complement colouring is balanced by parity. We claim balance, exactly; any connection of that balance to dynamical behaviour we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ, the fixed points of P_σ—the rules it maps to themselves—number 2^k, where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper's Figure 8 already exhibited a two-rule residual: a random field of tens of distinct rules dies off to an alternating population of just two, rules 42 and 170. Fittingly,

[3] `alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ, forcing each to number four.

the programming error that produced the family is what pins its residual at two—the erroneous mutation froze a bit, splitting the field into the cells that can reach 42 and those that can reach 170—so the phenomenon was on the page before the 2^k count explained it.

That die-off curve—the count of distinct rules present over time—is a natural population-level companion to λ. Where λ is a scalar assigned to a single rule, this curve is a property of the whole field's trajectory; and where λ is uninformative here—every fixed point sits at 1/2—the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 7).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Why Distance Geometry Cannot See This*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule shell—the field dies *onto the edge*, at $\lambda = 1/2$, always.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a single λ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 4 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R's four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that must collapse (Theorem 1) and that holds R among its fixed points. Carried out for Rule 54 this yields $\sigma = [2\,3\,0\,1\,5\,4\,7\,6]$; run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

The parity dichotomy in the spatial dynamics

Theorems 1–2 concern fixed points; run as a spatial dynamics, the parity dichotomy also shows in behaviour—but as a tendency, not a law. Only the all-even operators *can* settle onto a fixed rule, and some do (rotate+2), while others stay diverse though all-even (rotate+1); the any-odd operators, having no fixed rule, never settle (Figure 5). The full census below (*The full census, tabulated by class*) makes this quantitative over all 40,320 operators (Table 1): the dynamical content is the *death-rate difference* between the classes, not a clean collapse of the all-even class.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 5, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal aperiodicity.

The full census, tabulated by class

Definition 4 (Aliveness)**.** *A layer is* alive *on a run if its late-window* change rate*—the mean fraction of cells that differ between consecutive generations, over the last* 40 *of the* 100 *generations—is at least* 0.10*; otherwise it is* dead*. Genotype and phenotype are scored separately (three seeds), giving four regimes: live/live, live/dead, dead/live, dead/dead.*

The trend becomes a measured law once every operator is tabulated. We ran all 40,320 permutation-propagators and grouped them by cycle type (Table 1).

What the census shows

Cycle parity draws the line, exactly

Every cycle type with an *odd* cycle is 100.0% genotype-alive: having no fixed rule (Theorem 1), the field cannot collapse. Every *all-even* type splits—a fixed rule exists, but settling onto it is dynamical.

Collapse rises with the cycle count k

Among all-even types the alive fraction falls as the cycles shorten: [8] (k=1) is 83% alive, [2 6] and [4 4] (k=2) 70/67%, [2 2 4] (k=3) 49%, [2 2 2 2] (k=4) only 26%. This is the k-dependence the qualitative run could only suggest.

Most operators live; all four regimes are bijective

92.5% have a live genotype. The 2×2 split is 82.7% live/live, 9.8% live/dead, 6.0% dead/dead, 1.4% dead/live—so even Figure 3's live/dead mirror is reached by 3,953 *permutations*, not only by the non-bijective writings of Definition 1.

Table 1

Genotype-alive fraction by cycle type, over all 40,320 *permutations. Any odd cycle forbids a fixed rule and yields* 100% *life (Theorem 1); the all-even types split, collapse rising with the number of cycles* k.

cycle type	count	genotype-alive
any type with an odd cycle (17 types)	29,295	100.0%
[8]	5,040	82.9%
[2 6]	3,360	70.0%
[4 4]	1,260	66.7%
[2 2 4]	1,260	49.2%
[2 2 2 2]	105	25.7%

Fixed Points Are Not Attractors

Worked example 4: Reading the rotation survey

The fields of Figure 6 evolve under the *spatially coupled* dynamics of *Fixed Points Are Not Attractors*—the per-cell operator together with the coupling between cells—not P_σ applied cell-by-cell. This is what the survey turns on: under the bare operator every all-even rotation shown freezes onto its fixed rules (Figure 5, left). The coupling reverses this for the single 8-cycles alone—$\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3, fixed-point-bearing operators that nonetheless sustain a high-diversity field—while offset ± 2 (gcd 2) and ± 4 (gcd 4) collapse regardless; that split is what the survey shows.

The two panels are *almost* mirror images, and where they differ is instructive. Offset $+d$ and offset $-d$ are the same permutation exactly when $2d \equiv 0 \pmod 8$—so offset ± 4 (and the identity) are one operator with bit-identical panels, whereas offset ± 1 and ± 3 are distinct *inverse* rotations. Inverses share cycle structure, fixed rules, and the sustain-or-collapse verdict, but not their transient dynamics: offset $+1$ and offset -1 realise the same sustaining regime through visibly different fields.

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$. Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across 1/2, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Whatever pull toward balance the *spatial* field exhibits must therefore come from the coupling between cells and from the interrupter below, not from the bare map.

Empirical finding 1: The rotation survey

Every non-trivial wrap rotation is all-even and so possesses fixed rules (Theorem 1), yet in the spatially coupled dynamics they split by cycle structure (Figures 6, 7):

- the single 8-cycles—$\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3—*sustain* a high-diversity field, holding ∼30–40 distinct rules through the longest runs we made (400 generations);
- offset ± 2 ($\gcd 2$, two 4-cycles) and offset ± 4 ($\gcd 4$, four 2-cycles) *collapse* to one or two rules within a few dozen generations.

The distinct-rule population curve (Figure 7) separates the two by more than an order of magnitude in a single scalar. Possessing a fixed rule and having the field stay near it are different properties; only the second is visible in a drawing, and the interrupter (*Fixed Points Are Not Attractors*) is what holds the 8-cycles off their fixed rules. We do not claim the count is an order parameter in the statistical-mechanics sense; it is a summary of what the drawings show, on the two seeds per operator that we ran.

Definition 5 (Interrupter)**.** *An* interrupter *is a step that occasionally declines to propagate—a neighbour-blend that copies where neighbours agree, or a hold-when-active gate. Composed with the operator, the coupled dynamics no longer settles: instead of a* *frozen byte or a runaway cycle, the field holds a diverse population of rules, many at* $\lambda = 1/2$. *The coordinate-wise blend of Box 2 is the interrupter used in the ordinary spatial runs.*

Observation 3: Two couplings of one operator

This resolves the apparent tension between the census and Figure 7: they are the same operator under two *couplings*. Left to settle, the field under rotate+2 collapses onto a few of its fixed points; with the interrupter it is held off them and keeps tens of rules. Settling and sustained diversity are two behaviours of one operator, selected not by the bare algebra but by the coupling.

Observation 4: The family is flat under its own composition

Each operator of Equation (1) negates every bit ($P_\sigma(g)[j] = \neg\, g[\sigma^{-1}(j)]$, a coordinate permutation followed by a global flip), so under composition the flips cancel in pairs: $P_\sigma \circ P_\tau$ is a *pure* coordinate permutation with no negation left—a bijection that fixes 2^c bytes for c cycles but carries none of the forced $\lambda = 1/2$ structure. An even number of propagators composes to such a bare permutation, an odd number back to a single propagator; either way the composite contains nothing the individual operators did not. We searched composites of two and three operators for emergent structure and found none—the negation that gives one operator its balanced fixed point is exactly what a second cancels.

Propagators thus do not compose usefully with one another; the compositions that *do* yield something new must pair a propagator with a step of a different kind.

One such composition is worth exhibiting.

Definition 6 (River)**.** *A* river *is the composed operator obtained by following the collapsing offset-+2 propagator (*$\gcd 2$*) with a phenotype-reading construction step—match the local phenotype against four template candidates, first match wins, falling back to* *the all-zero rule—yielding a coherent band of alternating phenotype that drifts through a disordered background and reappears in all six seeds we tried (Figure 11).*

Where the bare rotation collapses or cycles, this composition holds an off-critical structure for the length of every run—its trajectory varying with the seed, its presence not. We report these two mechanisms here; a fuller catalogue of composed operators is future work.

Why Distance Geometry Cannot See This

Observation 5: Why per-trajectory metrics cannot see the classification

The classification is a conjugacy invariant of σ—its cycle type—while the complexity and compression measures we tried read a single rule's bits or one space–time trajectory. A quantity read off one field's history has no access to σ's cycle type, so it cannot recover the parity classification, and in our tests it did not: a population count reads a collapsed-but-maximally-selected field as nearly dead; a frozen periodic genotype scores a lower statistical complexity C_μ (Crutchfield & Young, 1989) than Rule 110; gzip rates a self-similar rule as noise. We claim no general impossibility theorem—only that the standard per-trajectory measures we tried do not separate the parity classes, unsurprisingly, because the classifying property lives on the operator, not on the states. The classification is a theorem *because* it is not a per-trajectory statistic.

Discussion

Is the surviving phase a critical point? We test and ask a sharper question

The survey (Figure 6) exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only the single 8-cycles (offset ±1, ±3) sustain a high-diversity phase, while offset ±2 and ±4 collapse. A reader is entitled to ask whether that phase sits at a *critical point*—tuned between an ordered and a disordered phase, as CA complexity is often supposed to (Langton, 1990). We test this directly with a finite-size scan and find no such point, so we ask a different, answerable question: whether the sustained diversity is spatially organised into coexisting dynamical regimes.

Empirical finding 2: No critical point: a drifting susceptibility peak, read as metastability

The natural control parameter is the *propagator fraction* q ($q = 0$ all blend, $q = 1$ all propagator, offset +1). Across widths $L = 30$–240 (32 seeds each) the genotype activity $a_G(q)$ rises *smoothly*, well described by a logistic crossover ($R^2 > 0.99$, Figure 12a); its width does not shrink with size ($w \approx 0.13$, $w \sim L^{-0.04}$), so the curve never steepens into the step a genuine transition would. The size-scaled run-to-run spread $L\,\mathrm{Var}(a_G)$ (Figure 12b) is a susceptibility: it *peaks* inside the crossover band, but the peak drifts in location ($q \approx 0.15$–0.5 across L) and grows only $\sim$1.1× from the smallest width to the largest—failing the convergence-and-growth test a critical point requires. At $q = 1$, the pure-propagator limit at which offset +1 runs, the spread is instead *low*: the field reliably organises into *system-spanning domains* (Figure 13), reproducibly rather than through a fluctuating transition. The honest reading is *metastability*, not criticality—seed-dependent residence times dominate this horizon, and we claim no confirmed critical point.

Empirical finding 3: A finite-width littoral edge, not a phase boundary

An intermediate-activity band ($0.15 \le a_G \le 0.60$) has a *lower* edge robustly at $q \approx 0.1$—stable under both system size and threshold—while its upper edge is threshold-dependent, because a_G saturates gradually rather than jumping (the band spans $q \approx 0.1$ to 0.4–0.5). This finite-width crossover is best read as a *littoral edge*: a shore of finite width rather than a thermodynamic phase boundary. Where the bare rotations collapse or cycle (Figure 6), offset +2 holds it for the length of the run.

Definition 7 (Isolated-rule activity score)**.** *The* activity score *of a rule is its asymptotic per-step cell-change rate when run alone as an elementary CA from a random initial* *condition—a run-time measure, distinct from the static truth-table balance* λ *(*Every Fixed Point Is Balanced*). Ordered rules score near* 0*; activity rises through the complex rules (Rule* 110 ≈ 0.4*) to the chaotic ones (*$\gtrsim 0.5$*). The score* orders *the regimes but does not by itself separate complex from chaotic.*

Tinting the genotype by this score (Figure 13) resolves the intermediate field into coexisting low- and high-activity domains whose boundaries trace filamentary sets through 1+1-dimensional spacetime—objects *analogous to* the domain boundaries of computational mechanics (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001), here obtained by thresholding a scalar tint rather than by predictive equivalence.

Empirical finding 4: The fractal domain walls are where the genotype is rewritten fastest

The activity field resolves into ordered and chaotic domains separated by *fractal-like walls* (Figure 14; box-counting dimension $D \approx 1.5$–1.8 over $L = 128$–768, a handful of seeds). Those walls are the locus of maximal *genotypic* change: measuring the row-to-row rewrite rate by region (Table 2), the wall churns more than either interior—for offset +2 at $L = 256$, 0.83 at the wall against 0.77 (ordered) and 0.58 (chaotic)—and exceeds *both* interiors in every one of the 32 committed operator/width/seed fields (offset +2, the river, $\sigma = 16250374$). The chaotic domain is in aggregate the most *genotype*-stable, so the wall—where ordered abuts chaotic—is where the propagator does its most rewriting. The walls are not compositionally special (no rule class enriched; Rule 110 if anything depleted), so the distinction is dynamical, not chemical. Whether that rewriting transports or combines information *along* the walls we do not test—the natural question for a sequel.

What we do not claim

Two stronger claims are deliberately withheld. We do not assert a critical transition—the scan is a null result, and the interface analysis is exploratory (few seeds; the fixed-boundary line variant rather than the wrapped lattice of Figure 6). Nor do we claim computation: held-out injection–decode tests validate genotype storage and limited transmission in the feedforward system, but under the river treatment the matched feedback contrast shows little capacity under these probes and no detectable XOR-like modification—and failure of these probes is not itself evidence of no computation. The denser river boundary ($D \approx 1.7$ versus 1.5) is an *unmatched* comparison and does not establish that the genotype–phenotype coupling causes the difference.

The sequel: do the filaments carry computation?

If the filaments are domain boundaries they are candidate *wires*. The natural question is not which rules compose them—in our data the complex rules are not enriched there—but whether information is *transported along* them and *combined where they collide* (Mitchell et al., 1993). Whether computation flows down the littoral filaments of these diagrams is a question for another paper.

Limits

The theorems are the Boolean case and are exact; the dynamical claims are narrower and empirical. The census and the sustained-diversity behaviour are reported for a single spatial coupling (S_8, radius one, one value map) at limited seed counts, and invite a fuller treatment. *Discussion* takes up the local, structural reading and reports an exploratory interface-dimension statistic under its own stated caveats (few seeds, fixed-boundary variant); beyond it we report no perturbation-spreading or entropy-rate statistics. Because the dynamics is stochastic, every such figure is given both as a fixed *representative* seed and, where a scalar is quoted, as a fixed *six-seed sample* (mean and range); both are reproducible from the pinned seeds through the accompanying package.

Table 2

Genotype churn by region—*row-to-row rewrite rate of the rule field (fraction of cells whose ECA rule changes from the row above), at $L = 256$, mean over the committed seeds. For every operator the wall exceeds* both *domain interiors; the chaotic interior—most active in phenotype—is the most genotype-stable. Wall extraction is the threshold-and-smooth procedure of Figure 14; reproducible from the committed fields (`plot_eoc_churn.py`).*

Operator	Wall	Ordered interior	Chaotic interior
offset +2	0.83	0.77	0.58
$\sigma = 16250374$	0.71	0.68	0.57
river	0.98	0.97	0.88

The substrate the object sits in is not itself new (*The Object*); λ is one heuristic among several (Wuensche, 1999), and we make no claim that balance implies any particular dynamics—only that here balance is *derived*. What is new, and what we rest the paper on, is the object of Equation (1) and the fixed-point structure that forces every fixed rule to be balanced, $\lambda = 1/2$, as a matter of proof.

Conclusion

The characterisation promised at the outset (*The Object*) has two halves. The first is static and exact: which operators possess fixed points (the all-even ones), how many (2^k), and where they lie (balanced, $\lambda = 1/2$, always). The second is empirical and tentative. Because every fixed point sits at $\lambda = 1/2$, that parameter is silent on the difference between an operator whose field stays diverse and one whose field collapses; the distinct-rule count is not, and in our runs it separates the two across the rotation family by more than an order of magnitude (Figure 7). The same observable offers a suggestive reading of the search that produced the family, reported both as a representative run and as a fixed six-seed sample (*Limits*). The operator of Figure 2 that comes to be dominated by Rule 110—a universal rule, the naive prize of a search for computational intelligence—concentrates the field onto it and pays for it in diversity. At the representative seed, 39% of its genotype cells are Rule 110 and it sustains 16 distinct rules, against a foil that holds Rule 110 below 1% and sustains 33; over the six-seed sample the ordering is unchanged—Rule 110 35% (range 31–39%) versus 0.2% (0–0.2%), sustained diversity 15 versus 23. On this pair, recovering a universal rule and sustaining a diverse population pull against each other—a suggestion worth testing at scale, not a law we have established.

Code and formal proofs. All code needed to reproduce the figures is available at `https://github.com/tothedarktowercame/mmca-clj` (commit `0163108ef96e180d79d921d6e90012a752b7830c`)—a standalone, dependency-light Clojure port of the MetaCA dynamics whose `README` gives the exact command for each figure. The classification and balance theorems are formally verified in Lean 4 with Mathlib at `https://github.com/tothedarktowercame/mathlib4`, branch `darktower` (commit `bc34d8cec27216aae1d3cd511a9950ab1bce3575`), in `DarkTower/Patterns/Propagator.lean`: the cycle-parity criterion as `hasAlternatingColouring_iff_cycleType_even` and the $\lambda = 1/2$ balance as `alternatingColouring_true_card_eq_four`.

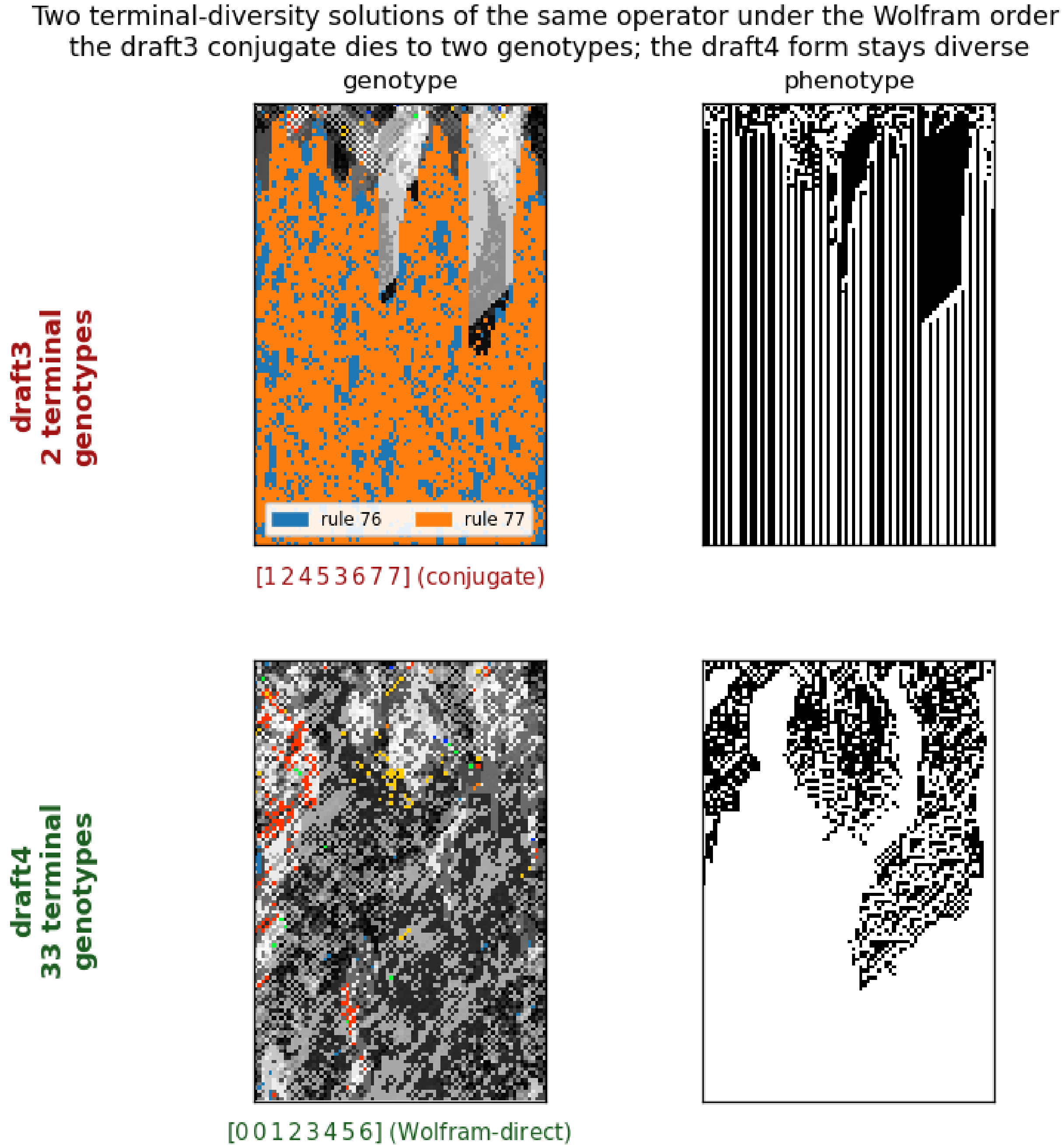

Figure 3

Two terminal-diversity solutions of one historical operator. *The historical write* [0 0 1 2 3 4 5 6] *has two forms under the standard Wolfram order. Its* conjugate [1 2 4 5 3 6 7 7] *(top) reproduces the draft3 behaviour: the genotype collapses to exactly two terminal rules—76 and 77, adjacent, differing in a single bit—regardless of seed, while the phenotype stays regular. Applied* directly *(bottom), the same operator keeps a diverse genotype (*∼30 *terminal rules)—the draft4 solution. Two ends of genotypic aliveness from one historical write.*

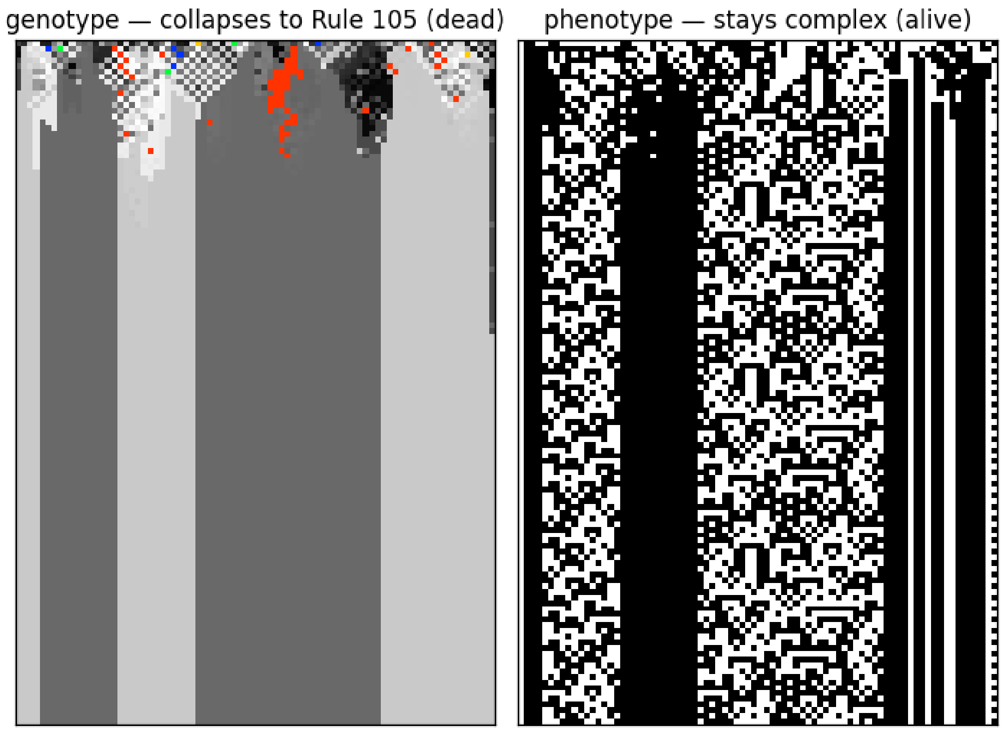

Figure 4

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [2\,3\,0\,1\,5\,4\,7\,6]$*, built to fix the balanced shell containing Rule* 105*, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule* 105*, an additive* $\lambda = 1/2$ *rule whose dynamics keep it complex. Seed* 1*,* $W = 80$*,* $T = 120$.

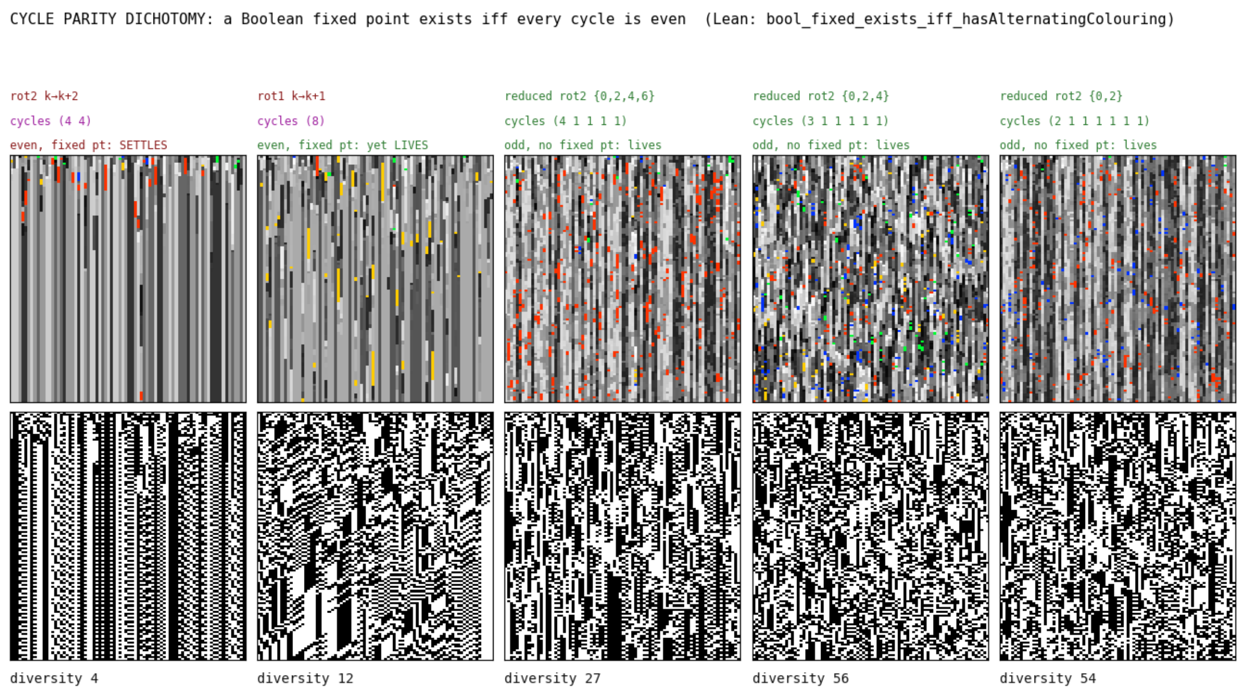

Figure 5

The parity dichotomy by example, illustrating Theorem 1 (a fixed rule exists iff every cycle is even). Five operators run alone, as genotype (top) and phenotype (bottom). Both all-even *operators possess fixed rules—but existence is not attraction: rotate+2 (cycles* 4 4*)* settles *onto its* $2^2 = 4$ *fixed rules (diversity* 4*, a regular phenotype), whereas rotate+1 (a single* 8*-cycle) does* not *settle—it stays diverse (diversity* 12*, an active phenotype), all-even yet alive. The three* any-odd *reduced operators possess no fixed rule at all, cannot settle, and stay richly diverse (diversity* 27–56*). That rotate+1 and rotate+2 share their parity but not their fate is the census in miniature (Table 1): all-even makes collapse* possible, *not inevitable.*

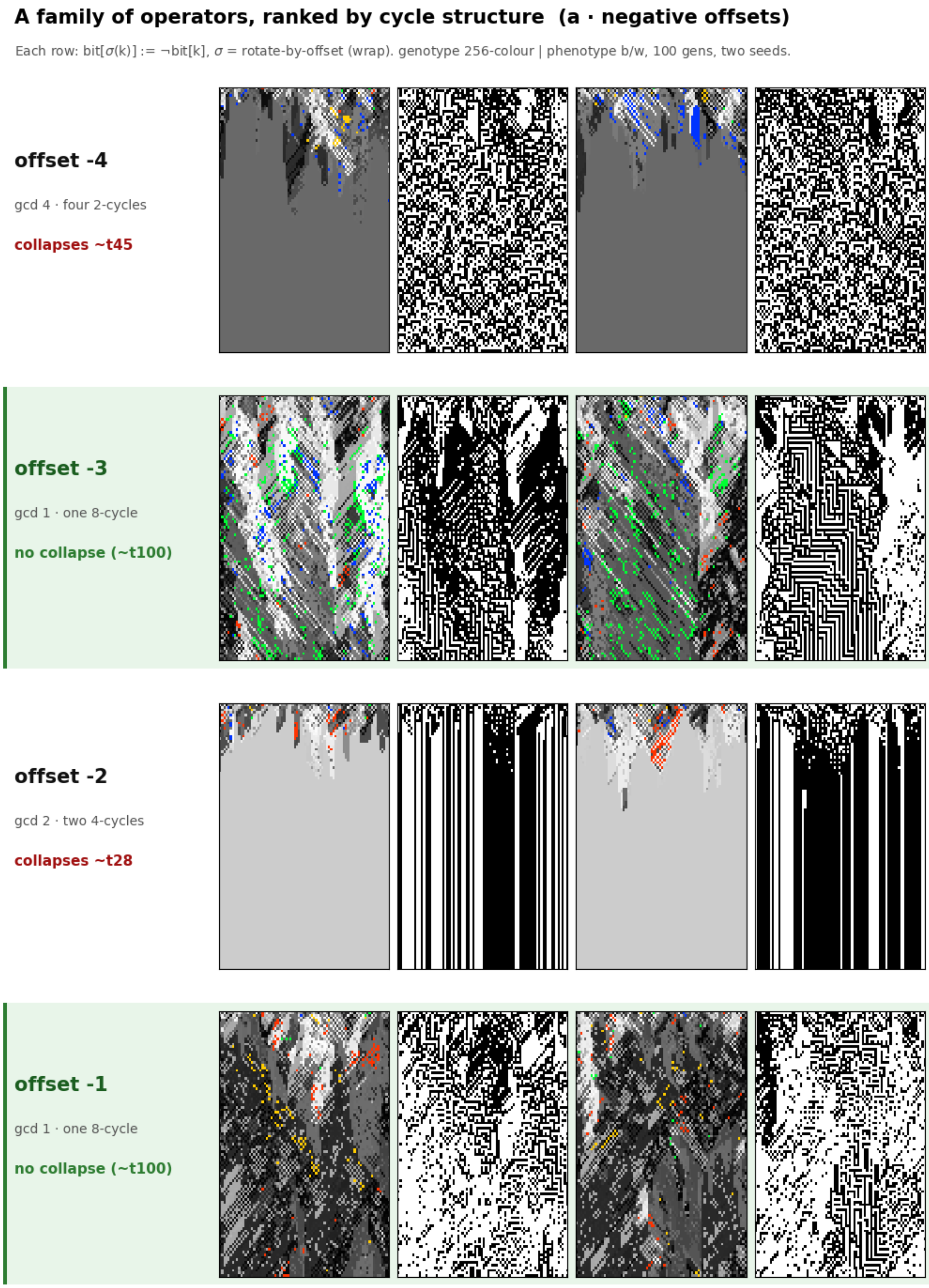

Figure 6

(a) Negative offsets. *The rotation subfamily, ordered by offset: each row is one wrap rotation* $\mathrm{bit}[\sigma(k)] := \neg\,\mathrm{bit}[k]$ *(two seeds; genotype in 256 colours, phenotype black and white, first hundred generations). Of these all-even rotations only the single 8-cycles,* $\gcd(\textit{offset}, 8) = 1$*—offset* ± 1 *and* ± 3*, highlighted—sustain a high-diversity phase; offset* ± 2 *(*$\gcd 2$*) and* ± 4 *(*$\gcd 4$*) collapse. See Worked Example 4; continued in (b).*

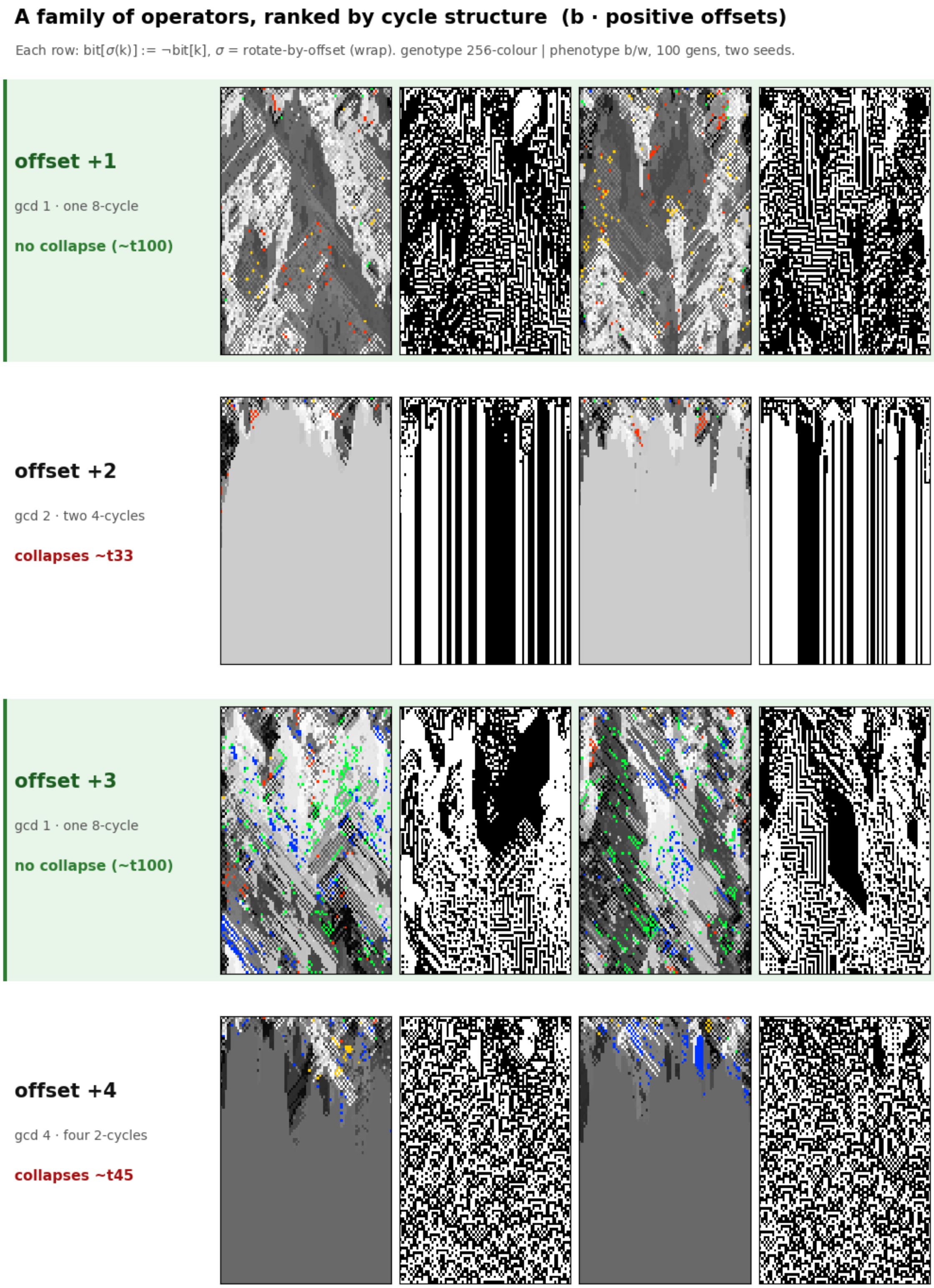

Figure 6

(b) Positive offsets. *Continued from (a). The classification is symmetric under* $d \to -d$ *(verdict follows* $\gcd(d, 8)$*), so offset* $+1$ *sustains as offset* -1 *does, and offset* $+2$ *collapses as offset* -2 *does—inverse rotations sharing verdict but not their transients (Worked Example 4). Possessing a fixed point and staying near it are different properties: this pair of pages is the whole paper in one figure.*

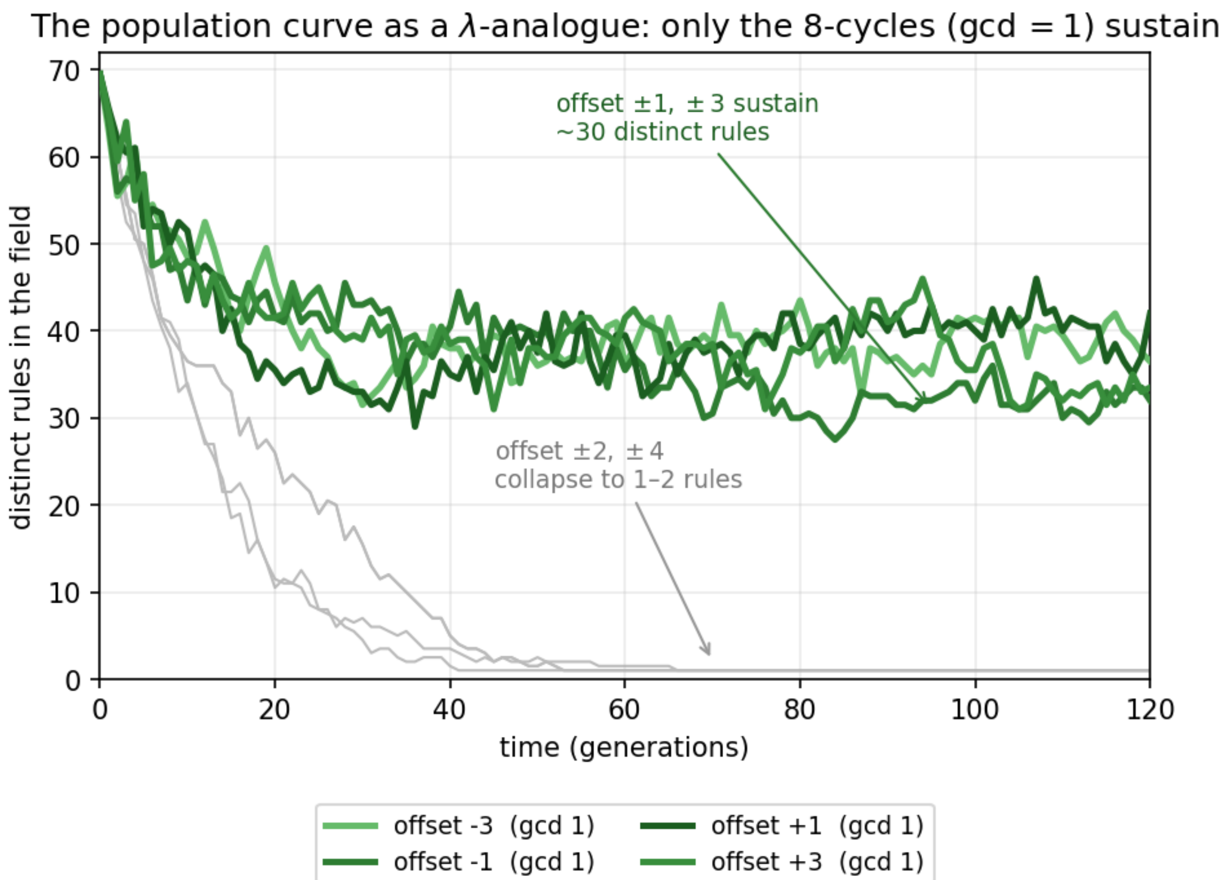

Figure 7

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70*; the four 8-cycles,* $\gcd(\textit{offset}, 8) = 1$ *(offset* ± 1 *and* ± 3*, green), keep a large diverse population, while offset* ± 2 *and* ± 4 *(grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure 6. The interrupting neighbour-blend is essential to the sustained case: without it these same operators, in the census runs, instead settle onto a handful of rules matching their* 2^k *count (the 8-cycles onto two, offset+2 onto four; Figure 5).*

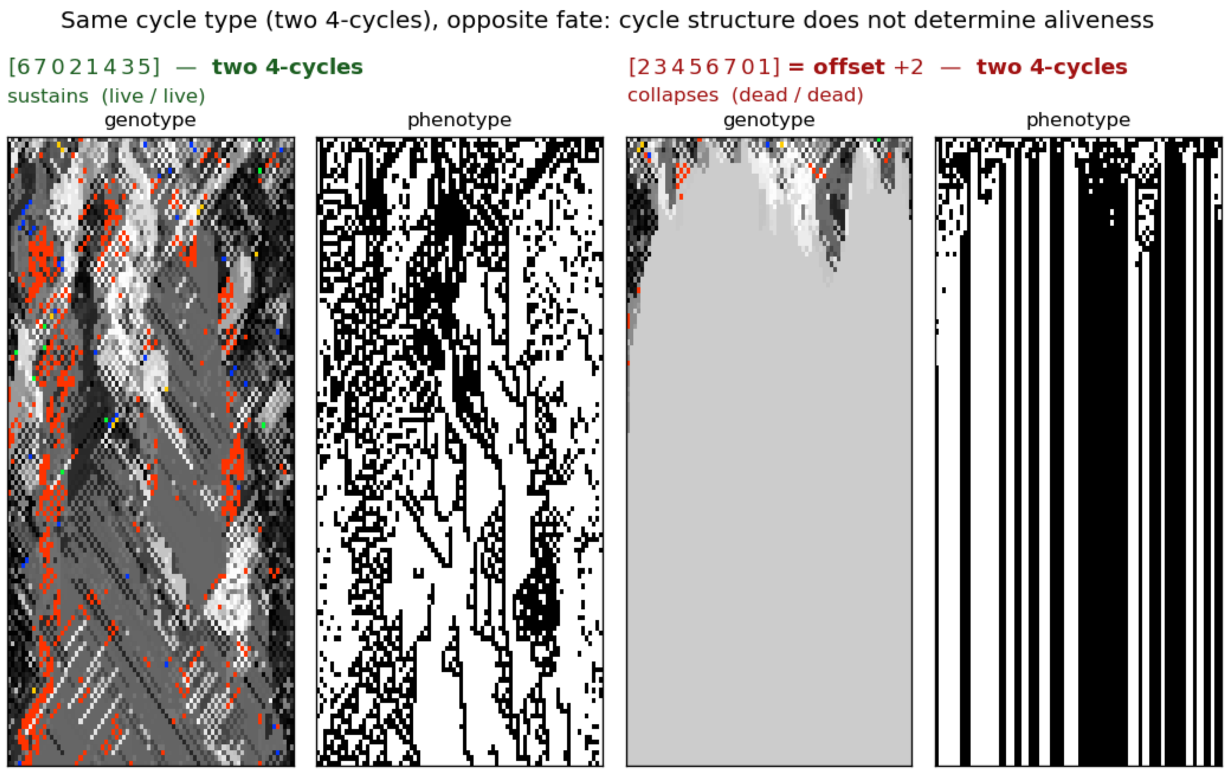

Figure 8

Cycle structure does not determine aliveness. *Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left,* [6 7 0 2 1 4 3 5] *sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the operator draft3 called offset+2 (its conjugate). Right, the Wolfram-order rotation offset+2* = [2 3 4 5 6 7 0 1] *has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.*

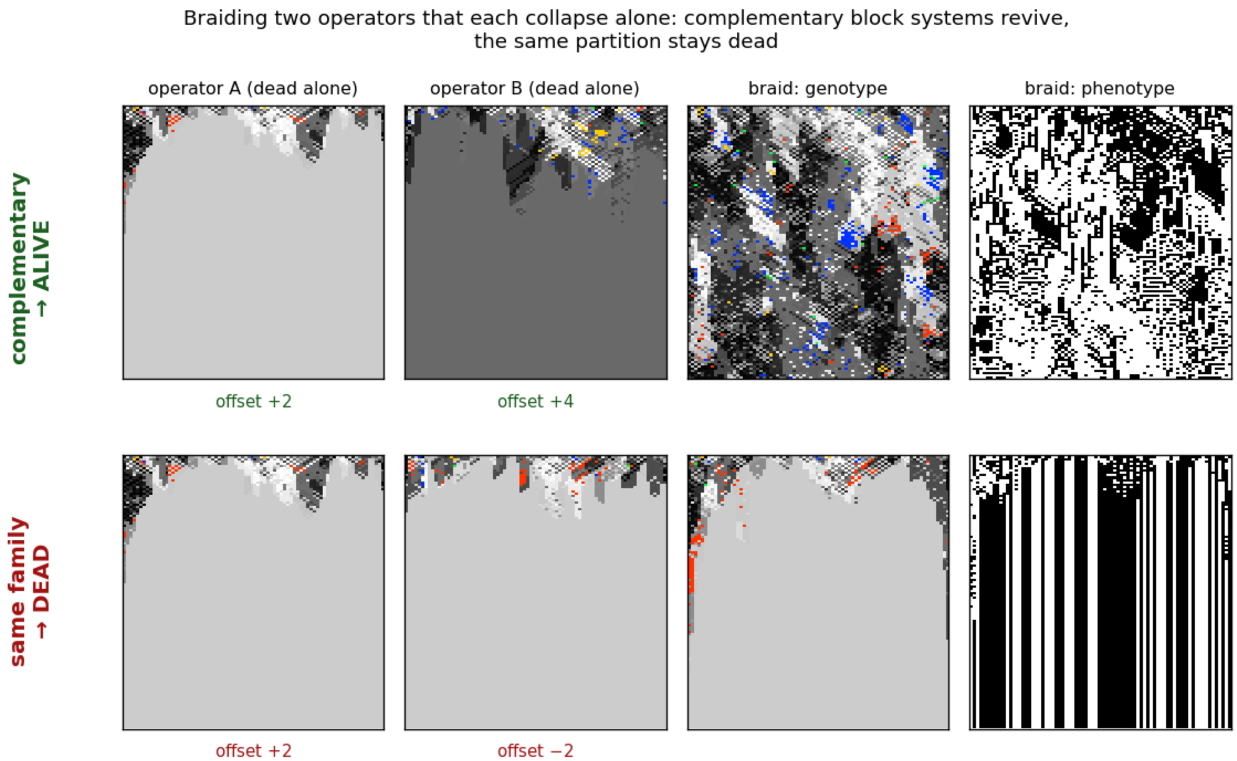

Figure 9

Braiding manufactures aliveness. *Two operators that each collapse alone can be* braided*—alternated step by step—into a sustained field, but only when they preserve* different *block systems. Top: offset*+2 *(two 4-cycles) and offset*+4 *(four 2-cycles) are each genotypically dead, yet their braid is alive (genotype change-rate* 0.76*, vs.* 0 *for either alone). Bottom: offset*+2 *braided with offset*−2 *stays dead—both preserve the same evens/odds partition, so the alternation never leaves it. Composition is thus a constructive tool: complementary dead operators synthesise a sustained, high-diversity braid.*

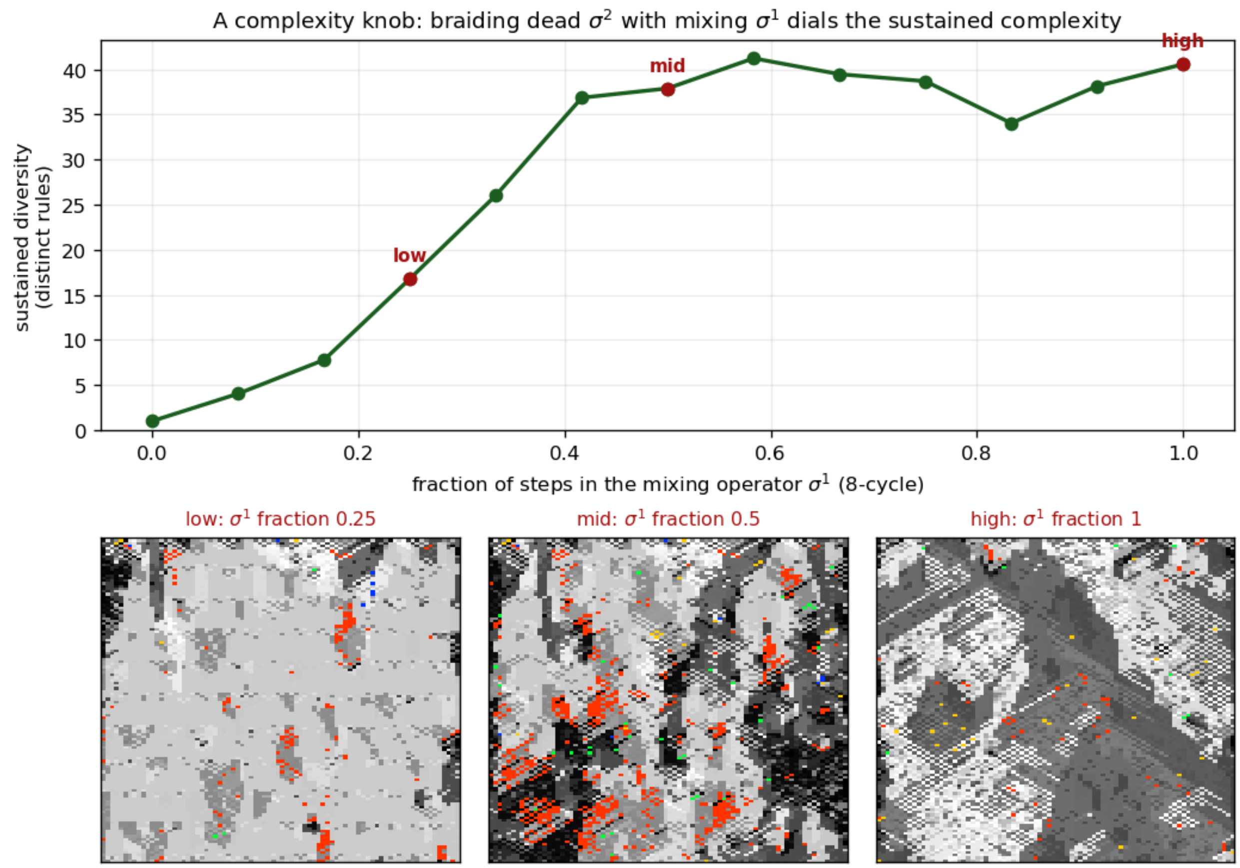

Figure 10

A complexity knob. *The powers of an 8-cycle σ form a lattice of block systems—σ^1 (no invariant partition, maximal mixing), σ^2 (two 4-cycles), σ^4 (four 2-cycles)—with the* intermediate *powers dead and the ends alive. Braiding the dead σ^2 with the mixing σ^1, the sustained diversity is set by the fraction of steps spent in σ^1: it climbs from dead (σ^1 fraction 0) to saturation (fraction $\gtrsim$0.4). The three panels show the field thickening from sparse (low) to a rich, high-diversity field (high) as the knob is turned—the same two operators throughout, only the schedule changing. Composition is thus not just constructive but* tunable*: a super-propagator that controls its own complexity.*

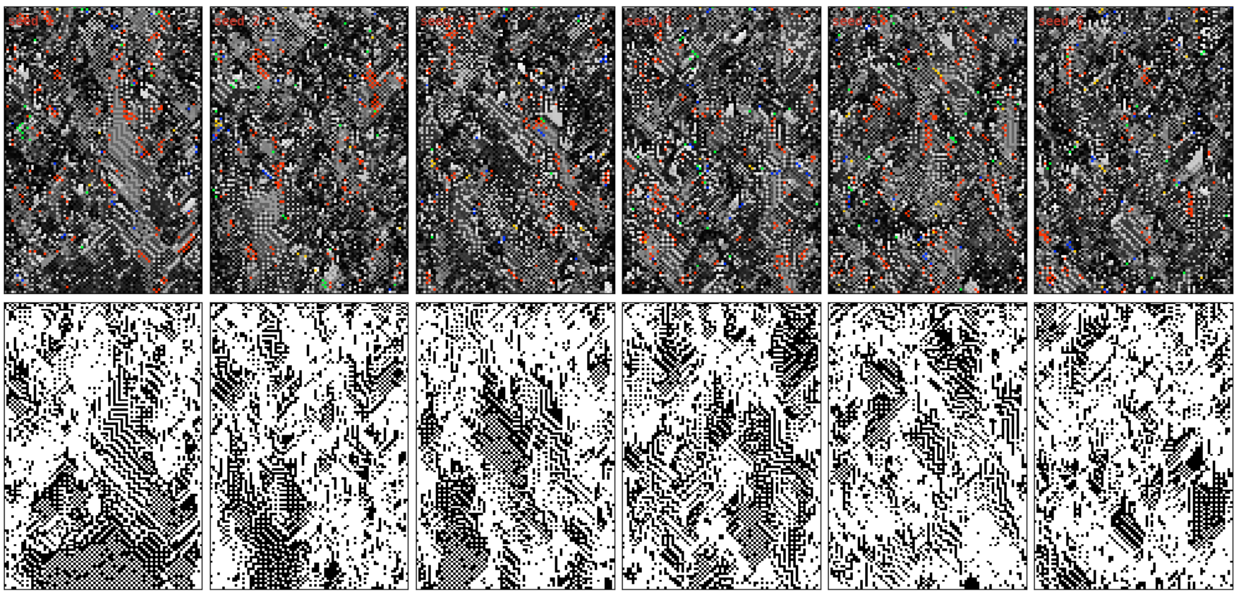

Figure 11

A river*: a composed operator sustains a reproducible structure. The offset-+2 propagator, composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule), produces in each of six seeds a coherent band of alternating phenotype (bottom row) drifting through a disordered background (genotype top, six seeds; genotype coloured by rule as in Figures 2–6—greyscale by byte value, with well-known ECAs highlighted, red = Rule 110). The band's trajectory varies with the seed; its presence does not. Where the bare rotations collapse or cycle (Figure 6), this composition holds an off-critical structure for the length of the run.*

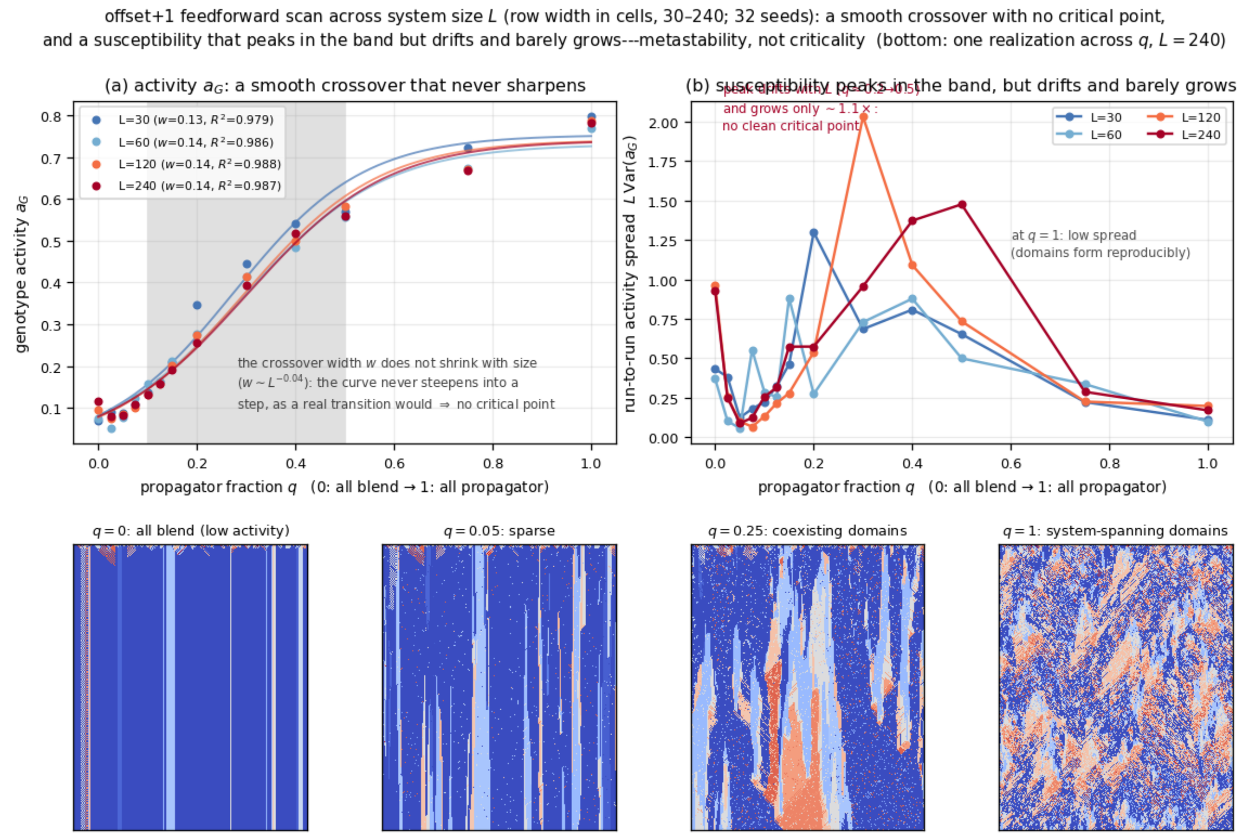

Figure 12

No critical point: the susceptibility peaks in the band but drifts and barely grows. *offset* $+1$ *feedforward finite-size scan against the propagator fraction* q *(*0*: all blend,* 1*: all propagator).* L *is the* system size*—the lattice width in cells; the scan spans* $L = 30$–240 *(32 seeds each), because a genuine transition sharpens as* L *grows while a crossover does not.* ***(a)*** *genotype activity* a_G *with logistic fits (*$R^2 > 0.99$*); the crossover width is size-independent (*$w \sim L^{-0.04}$*), so the curve never sharpens into a transition.* ***(b)*** *the size-scaled run-to-run spread* $L\,\mathrm{Var}(a_G)$ *is a susceptibility: it peaks inside the band, but the peak drifts (*$q \approx 0.15$–0.5 *across* L*) and grows only* $\sim 1.1\times$*, failing the convergence-and-growth test a critical point requires; at* $q = 1$ *the spread is low, the field forming system-spanning domains reproducibly. The honest reading is metastability, not criticality. Bottom: one realisation (*$L = 240$*, single seed) tinted by isolated-rule activity, from all-blend (*$q = 0$*) to the pure-propagator limit (*$q = 1$*).*

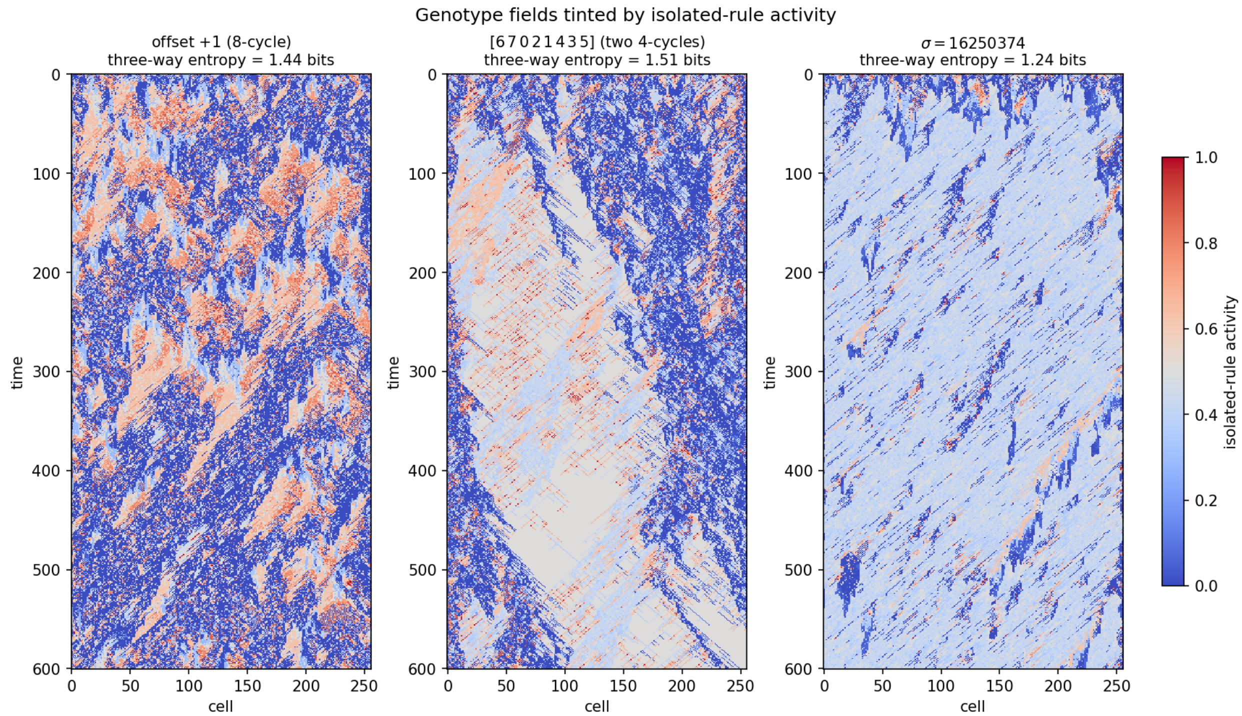

Figure 13

The genotype as a map of activity regimes. *Genotype spacetime fields (fixed-boundary variant) tinted by each rule's* isolated-rule activity score *(Definition 7; blue = low near* 0*, pale = intermediate near* 0.4*, red = high near* 0.5 *and above). Which active regime pairs with the low-activity background depends on the operator, and the three-way balance differs sharply (entropy in bits, in title): offset* +1 *(the 8-cycle survivor) mixes low with high-activity chaotic rules in a* fine *texture; the two-4-cycle* [6 7 0 2 1 4 3 5]*—the standard-order form of draft3's offset+*2 *exemplar—forms* large coherent *intermediate-activity domains with sharp walls, the most balanced regime mix (entropy* 1.51*); and $\sigma = 16250374$ is dominated by the intermediate complex Rule 110. The score orders the regimes but does not by itself separate complex from chaotic.*

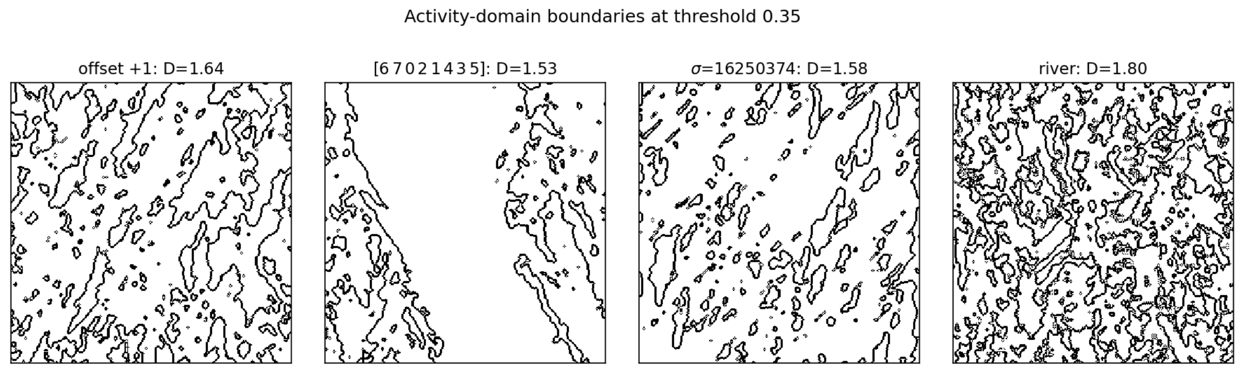

Figure 14

Activity-domain walls. *The boundary (black) between low- and high-activity regions, traced through the* 1+1*-dimensional spacetime diagram at* $L = 256$*. All the sustaining operators carry fractal-like boundaries (*$D \approx 1.5$–1.8*), but of visibly different density: offset* $+1$ *(the 8-cycle) and* $\sigma = 16250374$ *carry dense walls (*$D \approx 1.6$*), the phenotype-coupled river the densest (*$D \approx 1.80$*, an unmatched comparison as it couples phenotype), while the two-4-cycle* $[6\,7\,0\,2\,1\,4\,3\,5]$*—the standard-order form of draft3's offset*$+2$ *exemplar—carries* sparse, clean *walls bounding large single-activity regions (*$D \approx 1.53$*), matching its coherent domains in Figure 13. The dimension is stable across* $L = 128$–768 *(box sizes* 2–64*); collapsed operators (e.g. offset* $+4$*) carry none. These walls are also where the rule field is rewritten fastest (Emp. find. 4). Exploratory: a handful of seeds, fixed-boundary variant.*

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